Report of

“Intelligent MRI Diagnosis System” For

2CS501CC25

Machine Learning (ML)

B – Tech Semester IV

Report Prepared by



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1. INTRODUCTION

Brain tumors require accurate and timely diagnosis. This project presents an intelligent detection system using deep learning that not only classifies MRI images but also incorporates safety mechanisms and explainability features.

The system employs two approaches: Multi-Class Classification (identifying tumor type: Glioma, Meningioma, Pituitary, or No Tumor) and Binary Detection (presence/absence of tumor). Beyond classification, we introduce confidence-based decision making, image quality assessment, and explainable AI through Grad-CAM visualizations.

2. PROBLEM STATEMENT & OBJECTIVES

Challenges:

- Manual MRI analysis is time-consuming and requires expert radiologists
- Uncertainty in diagnosis can delay treatment
- Poor image quality leads to misdiagnosis
- Lack of explainability reduces clinician trust

Objectives:

1. Develop accurate models for brain tumor detection (target: >80% accuracy)
2. Implement confidence-based rejection for uncertain predictions
3. Add image quality assessment before prediction
4. Provide explainable AI through Grad-CAM
5. Build intuitive web interface for deployment

3. DATASET

Two datasets were used:

Multi-Class Dataset (Masoud Nickparvar - Kaggle):

- Total: 7,023 images (Training: 5,712, Testing: 1,311)
- Classes: Glioma, Meningioma, Pituitary, No Tumor
- Resolution: Preprocessed to 224x224

Binary Dataset (Navoneel Chakrabarty - Kaggle):

- Total: 253 images (155 tumor, 98 healthy)
- Used for cross-dataset validation
- Resolution: Preprocessed to 128x128

4. METHODOLOGY

4.1 Multi-Class Model (MobileNetV2)

Architecture: Transfer learning with MobileNetV2 pretrained on ImageNet
Configuration:
• Input: 224x224x3 RGB images
• Base: Frozen MobileNetV2 feature extractor
• Classifier: Custom fully-connected layers for 4 classes
• Training: Adam optimizer (lr=0.001), 10 epochs, batch size 16
• Augmentation: Random flips, rotations (±10°)
• Validation: 20% split from training data

4.2 Binary Model (Custom CNN)

Architecture: Improved custom CNN optimized for binary classification
Configuration:
• Input: 128x128x3 (smaller for faster inference)
• Layers: 4 conv blocks (32→64→128→256 filters) with BatchNorm
• Classifier: 512→128→2 neurons with Dropout (0.5, 0.3)
• Training: Adam optimizer (lr=0.001), 15 epochs, batch size 16
• Split: 70-15-15 (Train-Val-Test)

4.3 Novel Features

1. Confidence-Based Decision: Predictions with confidence <0.7 flagged for manual review

2. Image Quality Check: Pre-screening based on brightness analysis
   - Reject if brightness <30 (too dark) or >225 (overexposed)

3. Error Categorization: Systematic analysis of failure patterns
   - Categories: Low confidence, texture confusion, quality issues

4. Hybrid Decision Pipeline: Intelligent model selection based on use case
   - Binary for fast screening, Multi-class for detailed diagnosis

5. Grad-CAM Explainability: Heatmap visualization of model attention areas

5. IMPLEMENTATION

Technology Stack:

- Framework: PyTorch 2.10
- Libraries: NumPy, Pandas, OpenCV, Matplotlib, Seaborn
- Web Interface: Streamlit
- Development: Jupyter Notebooks

Project Structure:

- 3 Jupyter Notebooks: Training (multi-class, binary) and analysis
- 2 Utility modules: Custom dataset loader and Grad-CAM implementation
- Streamlit web app: User-friendly interface with real-time inference

Training Pipeline:

1. Data loading with custom PyTorch Dataset class
2. Train/validation split with augmentation
3. Model training with early stopping
4. Evaluation on separate test set
5. Novel features implementation

Computational Resources:

- Hardware: CPU-based (no GPU required)
- Training Time: 70 min (multi-class) + 20 min (binary)
- Inference: <2 seconds per image

6. RESULTS & ANALYSIS

Model Performance:

Multi-Class Model:
• Test Accuracy: 84.19%
• Training Time: ~70 minutes
• Parameters: 2.26M (31K trainable)

Binary Model:
• Test Accuracy: 82.14%
• Training Time: ~20 minutes
• Parameters: 4.98M (all trainable)

Cross-Dataset Validation:
• SoftMax Aggregation on Navoneel: 78.66%
• Demonstrates reasonable generalization across datasets

Confidence Analysis:
• High Confidence (>0.8): 67% of predictions
• Moderate (0.6-0.8): 24%
• Low (<0.6): 9% flagged for review

Error Analysis:
• Multi-class errors (15.81%): Mainly texture confusion between similar tumor types
• Binary errors (17.86%): Split between false positives and negatives
• Quality pre-screening prevents ~9% of errors

Metric	Multi-Class	Binary
Accuracy	84.19%	82.14%
Training Time	70 min	20 min
Parameters	2.26M	4.98M
Use Case	Type Classification	Detection

7. WEB INTERFACE

A Streamlit-based web application provides user-friendly access to the system:

Features:

• Model Selection: Choose between multi-class or binary detection
• File Upload: Drag-and-drop or browse for MRI images
• Quality Check: Automatic brightness and contrast analysis
• Real-time Prediction: Results displayed within 2 seconds
• Confidence Visualization: Color-coded confidence scores
  - Green (≥80%): Confident
  - Orange (60-80%): Moderate
  - Red (<60%): Low - manual review required
• Grad-CAM Overlay: Heatmap showing model focus areas
• Recommendations: Clear next steps based on results

User Workflow:

1. Select desired model type
2. Upload MRI image
3. System checks image quality
4. Click "Analyze" button
5. View prediction with confidence
6. Review Grad-CAM visualization
7. Follow recommendation

Deployment:

• Run locally with: streamlit run app.py
• Browser-based access at localhost:8501
• No technical knowledge required for users

8. NOVEL CONTRIBUTIONS

This project extends beyond standard classification:

1. System-Level Thinking: Rather than just training models, we designed a complete intelligent system addressing real-world deployment challenges.

2. Uncertainty Quantification: Confidence thresholding acknowledges when the model is uncertain, preventing overconfident misdiagnoses (9% of predictions flagged).

3. Quality Assurance: Pre-screening filters poor-quality images, preventing ~9% of potential errors before prediction.

4. Error Understanding: Systematic categorization reveals failure patterns, enabling targeted improvements.

5. Practical Deployment: Hybrid pipeline optimizes resources - binary for screening, multi-class for diagnosis.

6. Explainability: Integrated Grad-CAM builds clinician trust through transparent decision-making.

7. Cross-Dataset Validation: Testing on different dataset validates generalization (78.66% maintained).

These contributions transform a classification task into a trustworthy medical decision support system.

9. CHALLENGES & SOLUTIONS

Technical Challenges:

• Limited Dataset (Navoneel: 253 images): Solved with aggressive augmentation and transfer learning
• CPU-only Training: Chose efficient MobileNetV2 architecture
• Cross-Dataset Performance Gap: Acknowledged as validation, not failure

Implementation Challenges:

• PyTorch Setup: Created isolated virtual environment
• Grad-CAM Complexity: Modularized with extensive documentation
• Deployment: Streamlit simplified frontend development

10. CONCLUSION

This project successfully developed an intelligent brain tumor detection system achieving 84.19% multi-class and 82.14% binary accuracy. Beyond competitive performance, the system incorporates five novel features addressing real-world deployment challenges.

Key Achievements:
• Two trained models with strong performance
• Confidence-based safety mechanisms
• Quality pre-screening capabilities
• Explainable AI through Grad-CAM
• User-friendly web interface
• Cross-dataset validation

The system demonstrates that successful medical AI requires more than high accuracy - it needs uncertainty quantification, quality assurance, and explainability. By incorporating these features, we created a trustworthy decision support system suitable for clinical deployment.

Learning outcomes extend beyond technical skills to system-level thinking, addressing practical considerations for medical AI deployment.

11. FUTURE WORK

Potential Enhancements:
• Advanced architectures (EfficientNet, Vision Transformers) for 3-5% accuracy improvement
• Tumor segmentation using U-Net for precise boundary detection
• Multi-modal input combining T1, T2, FLAIR sequences
• Cloud deployment for remote access
• Mobile application for point-of-care use
• Clinical validation through prospective studies
• Integration with hospital PACS systems

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